8.2 Analyzing Arithmetic Sequences and Series

Name _____ Period ____ Date _____

Algebra II 8.2 · Student sheet

I can…

- tell whether a sequence is arithmetic and write its rule.
- write a rule from one term and the common difference, or from two terms.
- find the sum of an arithmetic series.

Vocabulary · fill in the missing word

Arithmetic sequence — the difference between _____ terms is constant

Common difference d — the _____ difference

Rule — a_n = _____

Arithmetic series — the sum of the terms of an arithmetic _____

Sum — S_n = _____

Part 1 · Gauss's trick

Write the sum forward and backward, add the columns, and count the pairs.

1 + 2 + 3 + … + 100 = _____ 1 + 2 + … + 40 = _____ 3 + 7 + 11 + … + 39 = _____

Part 2 · Notes: fill in the blanks

Arithmetic: the _____ between terms is constant (d).

Rule: a_n = _____

Sum: S_n = _____

Part 3 · Examples (work them with the class)

Example 1

Tell whether each sequence is arithmetic.

a. −9, −2, 5, 12, … b. 23, 15, 9, 5, …

Example 2

Write a rule for the nth term of 4, 9, 14, 19, … Then find a_{20}.

Example 3

Two terms of an arithmetic sequence are $a_4 = 20$ and $a_{13} = 56$. Write a rule for the nth term.

Example 4

Find Σ (i = 1 to 20) (3i + 7).

Example 5

The theater from the bell ringer has 15 rows: 20 seats in the first row and 2 more in each row. How many seats are in the theater?

Part 4 · Partner practice

Partner 1

One term of an arithmetic sequence is $a_6 = 32$, and $d = -5$. Write a rule for the nth term.

Partner 2

Find the sum of the first 15 terms of 2, 8, 14, 20, …